Reducing Patient No-Shows with Predictive AI: A Health Services Administration Case Study

Oct 25, 2026 · @Neha

1. Executive Summary & Problem Statement

Outpatient no-shows waste capacity a health system has already staffed and paid for, and a predictive model with targeted, equity-screened outreach can recover part of it, but only if outreach works well enough to cover its cost.

Problem statement. Missed appointments reduce access for other patients, delay care for the patient who did not attend, and leave fixed clinician and staff costs unrecovered. In one Veterans Health Administration analysis, about 18% of scheduled outpatient appointments in ten measured clinics went unused in fiscal year 2008, at an estimated cost of $564 million a year (Veterans Health Administration no-show study, PMC). The same paper put the cost of outpatient non-attendance to the UK National Health Service at about £790 million a year. A separate VHA cost study estimated an average of $196 lost per no-show and found that a centralized phone reminder lowered the no-show rate only from 16.3% to 15.8% (PMC4714455). Blanket reminders alone are not enough.

Approach. I built a no-show risk model on a synthetic dataset of 60,000 outpatient appointments, tested it on a held-out 30%, audited it for disparate impact, and costed a tiered intervention program under three outreach-effectiveness scenarios. All data is synthetic and labeled as such throughout; the structure and drivers follow published research, but the numbers are illustrative, not findings about real patients.

Headline results (synthetic data)

- **Targeting works.** The model reaches an AUC of 0.71. The riskiest 20% of appointments hold 42% of all no-shows and miss at a 39% rate, against 13.5% for the rest (2.1x lift). The top risk decile no-shows 46% of the time; the bottom decile, 6%.
- **Economics hinge on effect size.** Per 100,000 appointments a year, the program needs a roughly 19% relative reduction in flagged no-shows to break even. At a 9% reduction it loses about $122K; at 35% it nets about $191K.

- **Equity is a design constraint, not an afterthought.** Medicaid and self-pay patients are 30% of appointments but about 50% of high-risk flags. Dropping direct socioeconomic inputs cuts that to 38%, which is still above parity.

Recommendation. Deploy the model to drive patient-positive actions first: multi-channel reminders, two-way rescheduling, transport and telehealth offers, and waitlist backfill. Treat targeted overbooking as a later, fairness-gated step, and pilot before scaling so the real effect size is measured rather than assumed.

2. Methodology & Frameworks Applied

The project follows Lean Six Sigma's DMAIC cycle, uses a deliberately simple and auditable prediction approach, and layers equity and compliance checks on top, because a model that is accurate but unfair or non-compliant is not deployable in a health system.

Lean Six Sigma (DMAIC) mapping

Phase	What was done	Output
Define	Framed no-shows as lost capacity and delayed access; set the metric as no-show rate by clinic, specialty and lead time	Problem statement, target KPI
Measure	Built a 60,000-appointment baseline with an 18.1% no-show rate, comparable to the 6-19% range reported across published studies	Baseline dataset and rate by segment
Analyze	Ranked drivers by permutation importance: lead time, reminder confirmation and prior no-show history lead	Driver chart
Improve	Designed tiered interventions matched to risk and to the barrier behind it	Intervention playbook
Control	Defined monitoring: weekly no-show rate by tier, calibration drift, and flag-rate parity by payer group	Dashboard spec and review cadence

Predictive modeling approach

- **Models.** A logistic regression baseline and a gradient-boosted model, so the more complex model must earn its place. They performed almost identically (AUC 0.710 vs 0.705), which is a useful finding: the simpler, more explainable model is the better operational choice here.
- **Validation.** Hold-out split (first 70% of records to train, last 30% to test; a real deployment should split by date), with AUC, precision-recall AUC, Brier score, calibration by risk decile and lift as the measures. Published no-show models typically land in this range; one dental-clinic study reported an AUC of 0.718 (PMC9680883).
- **Use of the score.** The score ranks appointments for action; it is not a verdict on a patient. Thresholds are set by outreach capacity (here, the top 20%).

Equity and AI governance

- **Patient-positive first.** An academic health system that piloted a vendor-built EHR no-show predictor chose to act only on interventions that help the patient, because overbooking risked diverting resources from vulnerable patients (Health Affairs Forefront). That pilot's targeted outreach in 12 clinics cut no-show rates by a mean of 9% versus the prior year, which is why 9% is the conservative scenario in this model.
- **Proxy bias.** Removing race or insurance does not remove bias, because prior no-shows track socioeconomic barriers. A peer-reviewed study found Black patients were disproportionately placed in overbooked slots by a cost-minimizing scheduler, and showed a race-aware objective removed the disparity without a meaningful cost increase (Samorani et al., SSRN). This project therefore audits flag rates by payer group rather than assuming neutrality.

Regulatory and compliance considerations

These are the frameworks a real deployment would have to clear. They are listed for scoping and should be confirmed with compliance counsel before use, since rules and enforcement status change.

- **HIPAA.** Reminder and rescheduling outreach is a treatment and health care operations use of protected health information. It needs minimum-necessary data handling and a business associate agreement with any AI or messaging vendor.
- **TCPA and consent.** Automated texts and voice calls require documented patient consent and opt-out handling.

- **ACA Section 1557 nondiscrimination.** HHS's 2024 rule added obligations to identify and mitigate discrimination risk in patient care decision support tools. Confirm current effective dates and scope.
- **Medicaid managed care access standards.** Appointment wait-time and network adequacy standards give a regulatory reason to reduce wasted capacity, particularly for Medicaid populations.
- **Language access and disability access.** Outreach must be available in patients' preferred languages and accessible formats.

3. Mock Data & Strategic Interventions

On a synthetic 60,000-appointment dataset, a risk model ranks appointments well enough to target the top 20% for outreach, and the intervention plan is matched to risk tier and to the barrier behind the risk.

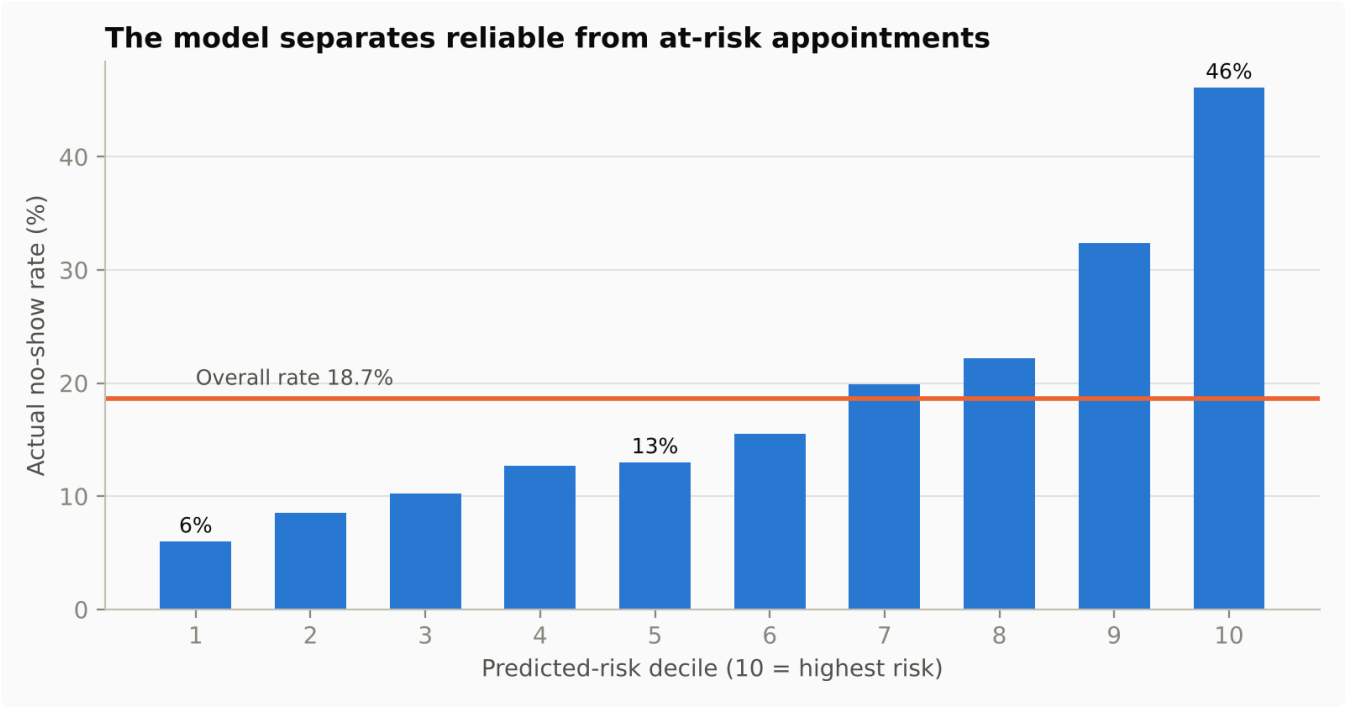
Dataset (synthetic)

Each record is one scheduled outpatient visit with lead time, new-patient status, age, specialty, payer, a neighborhood deprivation index, a transportation-barrier flag, day and hour, whether the reminder was confirmed, and smoothed prior no-show history. Patient propensity to miss is generated as an unobserved trait correlated with deprivation, so models can recover it only through observed history. The overall no-show rate is 18.1%. The generating code and data are in `noshow_analysis.py` and `appointments_synthetic.csv`, so every number here is reproducible.

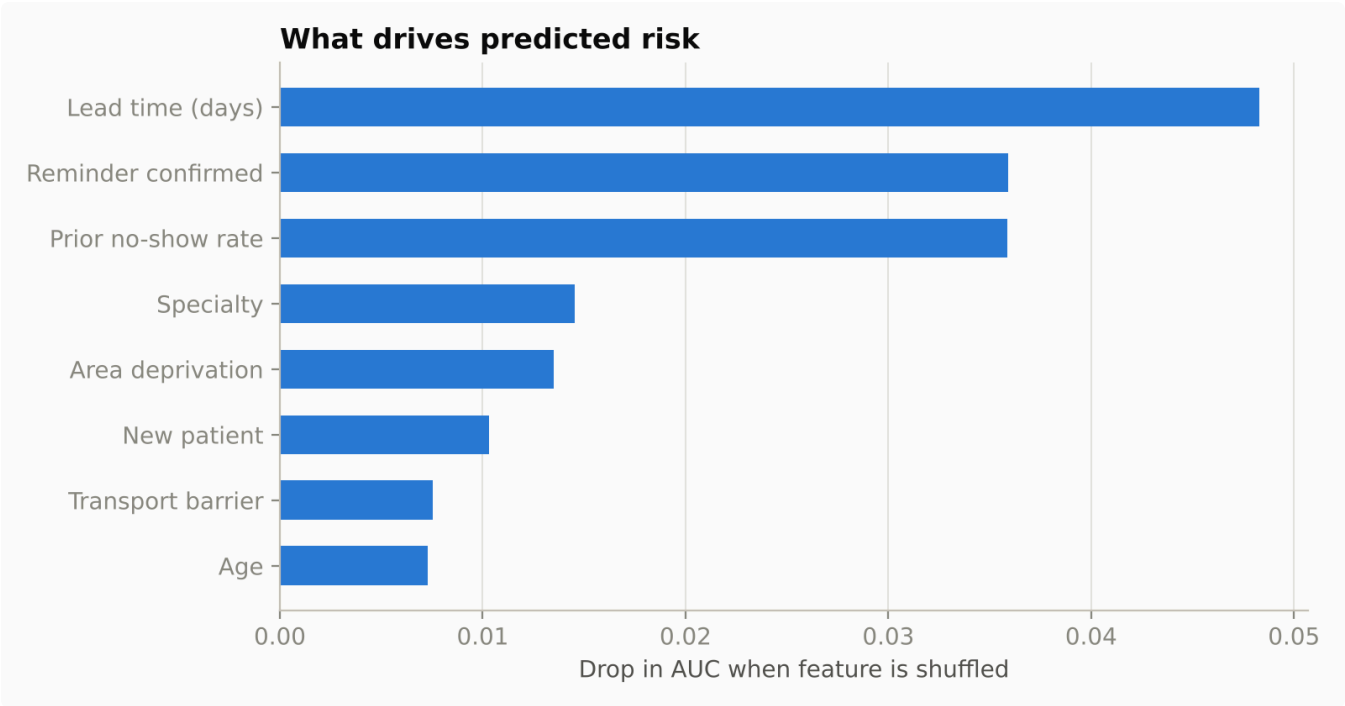
Model performance (held-out 18,000 appointments)

Model	AUC	Precision-recall AUC	Brier score
Logistic regression	0.710	0.395	0.1359
Gradient boosting	0.705	0.385	0.1369
Gradient boosting, no socioeconomic inputs	0.696	n/a	n/a

The base-rate precision-recall AUC is 0.187, so both models roughly double it. Predicted and actual rates track closely in most deciles; the largest gap is about 4 points in decile 9, where the model under-predicts, so scores should be recalibrated before being read as literal probabilities.



Lead time, reminder confirmation and prior no-show history are the strongest drivers, which matches what published studies report and means the levers are actionable: shorter booking windows and confirmed reminders.



Tiered intervention playbook

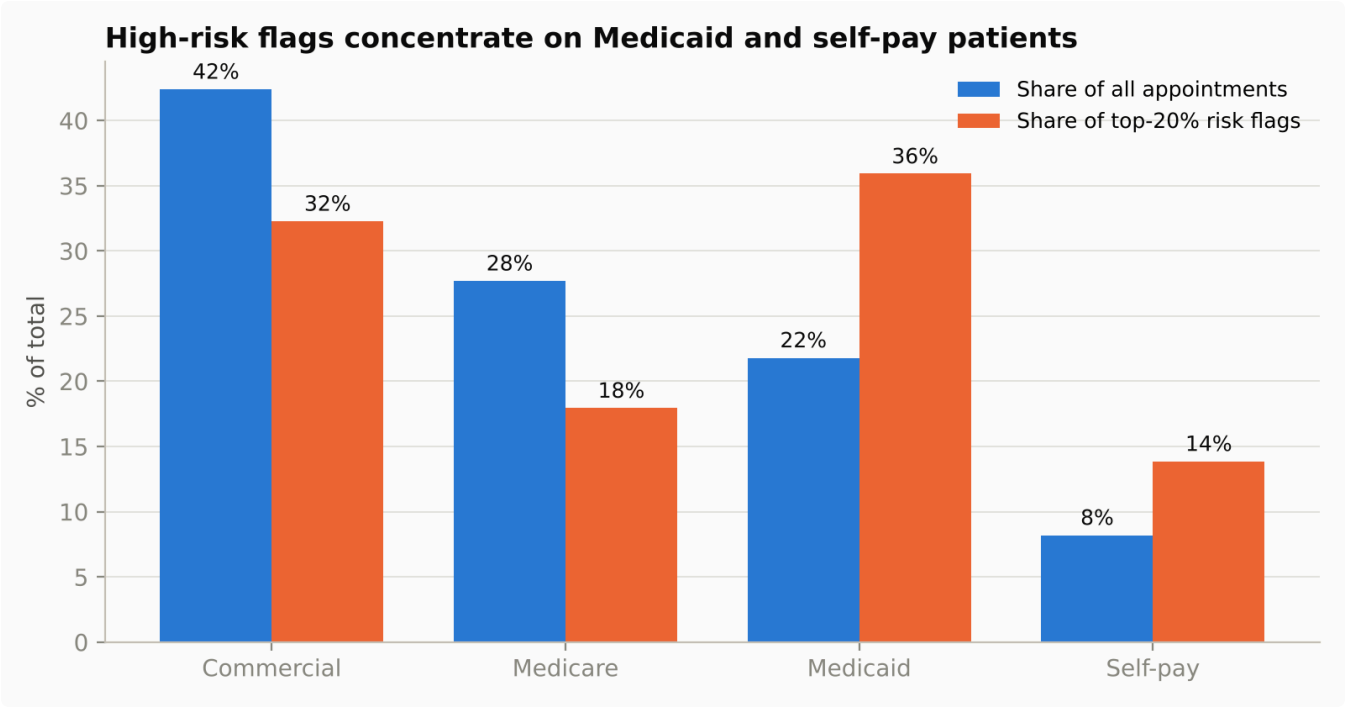
Tier	Risk deciles	Share of visits	No-show rate	Interventions
Low	1-6	60%	11.0%	Standard automated reminder; self-service reschedule link
Moderate	7-8	20%	21.1%	Two-way text with confirm, cancel or reschedule; second touch 48 hours out
High	9-10	20%	39.3%	Multi-touch AI voice and text; staff call for unconfirmed visits; barrier screening with transport, telehealth or flexible-time offers; waitlist backfill the moment a slot opens

The high tier holds 42% of all no-shows, so concentrating the costlier touches there is the efficiency case for prediction over blanket outreach.

Fairness audit

Medicaid and self-pay patients are 30% of appointments but about 50% of high-risk flags. A model with direct socioeconomic inputs removed still flags them at 38%, because prior no-shows carry the same signal.

Payer group	Share of visits	Actual no-show rate	High-risk flag rate	Flag rate, no socioeconomic inputs	Share of no-shows caught
Commercial	42.4%	16.3%	15.2%	20.4%	35.2%
Medicare	27.7%	16.7%	13.0%	13.8%	29.6%
Medicaid	21.8%	23.8%	33.0%	24.9%	55.7%
Self-pay	8.2%	24.0%	33.8%	25.9%	59.6%



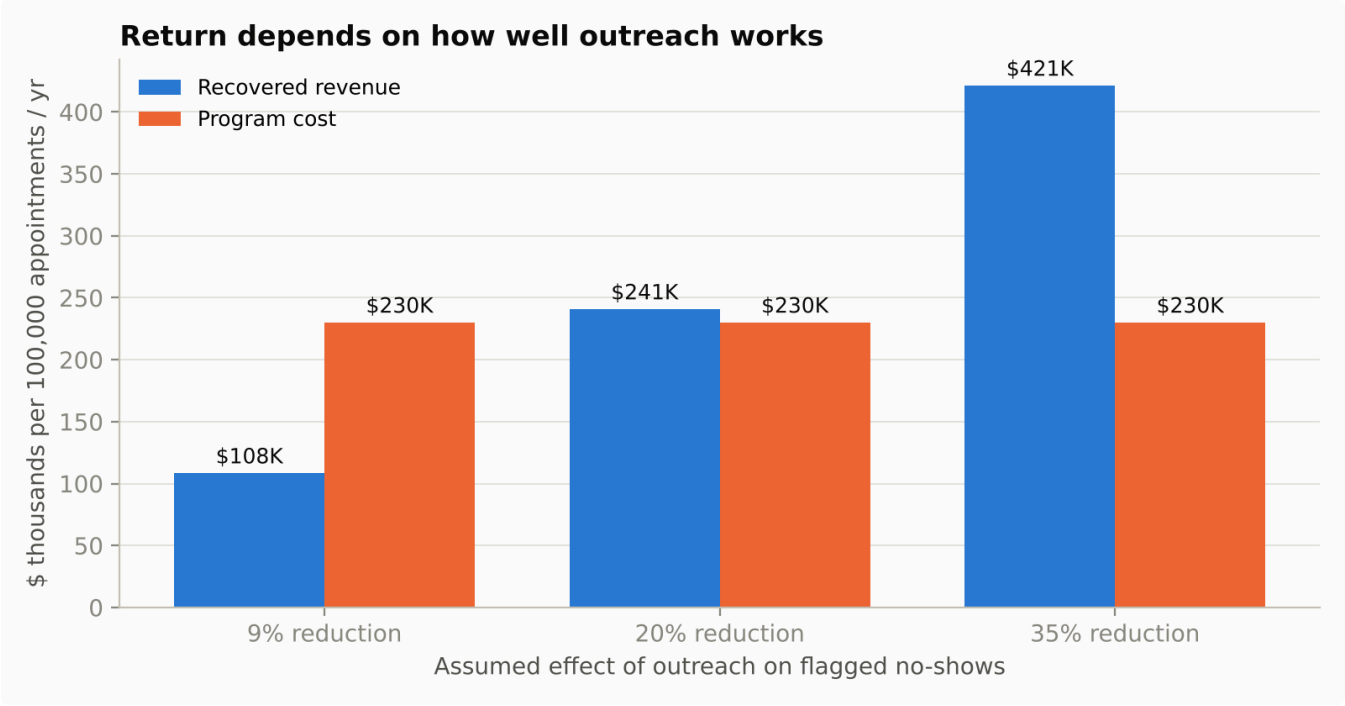
The implication is a policy, not a model tweak: use flags to give vulnerable patients more help, never to give them worse slots. Overbooking, if ever added, should be gated on parity checks and a fairness-aware objective.

Financial model (per 100,000 appointments a year)

All inputs are assumptions I chose and can be changed; none are measured results.

Assumption	Value
Baseline no-shows	about 18,700
Appointments flagged (top 20%)	20,000, holding about 7,850 no-shows
Value of an empty slot recovered	$200 (literature reports roughly $196 to $200 per missed visit)
Outreach cost per flagged appointment	$4.00
Platform, integration and analytics support	$150,000 a year
Share of prevented no-shows that attend vs cancel early	2/3 attend, 1/3 cancel early
Early-cancelled slots refilled from waitlist	30%

Outreach effect on flagged no-shows	No-shows prevented	Slots recovered	Gross benefit	Total cost	Net	Return per $1
9% reduction (Health Affairs pilot)	706	542	$108K	$230K	-$122K	$0.47
20% reduction	1,570	1,204	$241K	$230K	+$11K	$1.05
35% reduction (upper bound)	2,748	2,106	$421K	$230K	+$191K	$1.83



Break-even needs about a 19% reduction in flagged no-shows. Vendors report reductions of 20% to 40%, but those claims come from vendor marketing and are not independently verified, so I treat 35% as an upper bound and recommend a measured pilot before committing to the platform cost. The model also omits benefits (shorter waits, continuity of care, staff time) and costs (overbooking collisions), so it understates both.

4. Key Deliverables & Outcomes

The project delivers a risk model, a tiered outreach playbook, a fairness-gated governance process and a phased rollout, with a mock outcome target of an 8% relative reduction in the overall no-show rate in the base case and 15% as a stretch, each tied to a stated outreach-effect assumption.

Mock outcome targets (12 months, pilot clinics)

The overall reduction follows directly from the financial model: outreach effect on flagged no-shows, applied to the 42% of no-shows the high-risk tier holds.

Metric	Baseline (synthetic, held-out period)	Base case (20% outreach effect)	Stretch (35% outreach effect)
Overall no-show rate	18.7%	17.1% (8.4% relative reduction)	15.9% (14.7% relative reduction)
High-risk tier no-show rate	39.3%	31.4%	25.5%
No-shows prevented per 100,000 visits	n/a	1,570	2,748
Net financial impact per 100,000 visits	n/a	+$11K	+$191K
Share of high-risk flags held by Medicaid and self-pay	about 50%	40% or lower	35% or lower

The 15% figure is only reachable at the upper-bound effect size. Presenting both cases, and showing that the 9% case loses money, is deliberate: it shows a recruiter that the numbers were stress-tested rather than reverse-engineered.

Phased rollout with decision gates

Phase	Timing	Key activities	Gate to proceed
1. Discovery	Months 0-2	Audit scheduling and EHR data quality; set clinic-level baselines; complete privacy, consent and vendor agreements	Baseline signed off by operations; compliance clearance

Phase	Timing	Key activities	Gate to proceed
2. Silent pilot	Months 3-4	Run the model in the background in 3 clinics; compare predictions with actual attendance; audit flag rates by payer group	Calibration reviewed per decile and recalibrated where gaps exceed 3 points; flag parity reviewed
3. Live pilot	Months 5-9	Start tiered outreach in pilot clinics with matched control clinics	Measured reduction of at least 19% on flagged visits (break-even) and no worsening of parity
4. Scale	Months 10-14	Extend to remaining primary care and high-volume specialties; automate waitlist backfill	Return above $1 per $1 sustained for two quarters
5. Optimize	Ongoing	Quarterly retraining and drift checks; evaluate fairness-gated overbooking only if equity metrics hold	Governance committee approval

Governance and ownership

Ambulatory operations owns outcomes, analytics owns the model and its monitoring, and compliance and a patient-equity representative review fairness and consent each quarter. Any change to how flags are used, especially anything touching slot allocation, needs committee sign-off.

Principal risks

Risk	Mitigation
Outreach works less well than assumed	Gate scale-up on measured pilot effect against the 19% break-even
Model drifts as patient mix or access patterns change	Quarterly recalibration; alert if any decile drifts more than 3 points from its launch calibration

Risk	Mitigation
Disparate impact on vulnerable patients	Patient-positive actions first; parity monitoring by payer, language and neighborhood
Overbooking collisions raise waits	Defer overbooking; use waitlist backfill instead
Consent or privacy gaps in automated outreach	Documented consent and opt-out; business associate agreement; minimum-necessary data
Staff do not trust or use the flags	Show the drivers behind each score; train front-desk teams; collect feedback monthly

5. Visual Assets & Artifacts to Build

Build six assets, in this order of recruiter impact: an executive deck, a dashboard mockup, a one-page brief, a patient-journey process map, a code repository, and a front-line playbook.

#	Asset	Format	What it shows	Why a recruiter cares
1	Executive slide deck	10-12 slides	Problem and cost, DMAIC approach, model results, tiered playbook, fairness audit, three-scenario ROI, rollout gates, ask	Shows you can brief a C-suite in a decision format
2	Predictive operational dashboard mockup	Interactive page or screenshots	KPI tiles (no-show rate, high-risk tier rate, slots recovered), risk-tier funnel, payer-parity panel, ROI scenario selector	Shows operational analytics and monitoring, the Control phase of DMAIC
3	One-page executive brief	PDF	Problem, three headline numbers, recommendation, key risk	The asset most likely to be read in full

#	Asset	Format	What it shows	Why a recruiter cares
4	Patient-journey process map	Swimlane diagram	Booking, reminders, confirmation, risk flag, outreach, reschedule, waitlist backfill, with owners and AI touchpoints	Shows process redesign, not only analytics
5	Code repository	GitHub with README	`noshow_analysis.py`, synthetic data generator, results and charts	Proves the numbers are reproducible
6	Front-desk intervention playbook	1-2 page SOP	What each risk tier triggers and who does what	Shows implementation thinking, not just strategy

Suggested 11-slide deck outline

1. Title and one-sentence answer
2. The cost and access problem, with sourced figures
3. Project scope and approach (DMAIC)
4. What drives no-shows
5. Model performance and lift by risk decile
6. Tiered intervention playbook
7. Fairness audit and governance guardrails
8. Financial model and break-even
9. Outcome targets: base case and stretch
10. Rollout with decision gates
11. Risks, limits and next steps

Presenting it to recruiters

- Lead with the business question and the recommendation, then show the method.
- Say "synthetic data" on the first slide and in the README; it signals rigor, not weakness.

- Show the 9% scenario that loses money. It is the clearest evidence of judgment in the project.
- Prepare a 60-second answer to "What would you do differently with real data?" (measure the true effect with a matched-control pilot, validate drivers locally, and audit fairness by language and neighborhood as well as payer).

Key Takeaways

Prediction makes outreach efficient, but the program is worth running only if outreach works well enough and is applied fairly.

1. **Targeting beats blanket outreach.** The riskiest 20% of visits hold 42% of no-shows, a 2.1x lift, so the costliest touches can go where they matter most.
2. **The economics hinge on effect size.** Break-even needs about a 19% reduction in flagged no-shows. Per 100,000 visits a year, a 9% effect loses about $122K, a 20% effect nets about $11K, and a 35% effect nets about $191K.
3. **Two of the top three drivers are controllable.** Lead time and reminder confirmation, alongside prior no-show history, drive risk, so shorter booking windows and confirmed reminders are direct levers.
4. **Equity is a design constraint.** Medicaid and self-pay patients are 30% of visits but about 50% of high-risk flags, and 38% even without direct socioeconomic inputs. Use flags to add help, and defer overbooking until parity checks pass.
5. **Pilot before scaling.** A silent pilot, then a live pilot with matched control clinics, with scale-up gated on a measured effect of at least 19% and two quarters above $1 returned per $1 spent.
6. **Treat the numbers as illustrative.** All results use synthetic data and assumed effect sizes, so the value of the project is the method and the judgment, not the specific figures.

Limitations and Sources

The quantitative results are illustrative because the data is synthetic, and the research figures below come from search-result excerpts that should be checked against the full papers before you quote them in an interview.

Limitations

- Synthetic data reproduces known drivers by design, so model accuracy and lift show the method works, not what a real health system would see.
- Outreach effect sizes are assumptions. The 9% figure comes from one pilot; the 20% and 35% scenarios are hypothetical, and vendor-reported reductions of 20% to 40% are unverified marketing claims.
- The financial model values a recovered slot at a flat $200 and ignores payer mix, visit type, overbooking collisions and patient-experience effects.
- The study cites are from different decades and settings (VHA, NHS, academic clinics); no-show rates and costs vary widely by specialty and region.
- Regulatory items are a scoping checklist, not legal advice; confirm current rules with compliance counsel.

Sources

- VHA no-show variation and cost study (PMC4934549): 18% unused appointments, $564M a year, and NHS £790M a year
- Prevalence, predictors and economic consequences of no-shows (PMC4714455): $196 average cost per no-show; modest effect of phone reminders
- Predicting dental no-shows (PMC9680883): example AUC of 0.718
- Predicting no-shows in a rural health system (PMC10503036): modeling approach for overbooking
- Decision analysis framework for predicting no-shows, BMC Health Services Research (2024): handling imbalanced data
- Discrimination by AI in a commercial EHR, Health Affairs: patient-positive interventions and a 12-clinic outreach pilot with a 9% mean reduction
- Overbooked and Overlooked, Samorani et al. (2021): racial bias in no-show-based overbooking
- Project files: `noshow_analysis.py`, `appointments_synthetic.csv`, `results.json` (synthetic data; all figures in this case study)